Transmitter Audio Input Verification and Switching Tones Using BMH

Transmitter Audio Input Verification (Transmitter Site)

1. If not already done, call the appropriate WFO to request that they send a PNS stating that the transmitter will be taken off the air for maintenance.

2. Set up the IFR 1200S (or 1200SS) Communication Monitor, using Attachment D.

3. Using clip leads, connect the AM-48 Test Set (refer to Attachment C for setup) or RMS Voltmeter in parallel to the transmitter telephone line audio input in the back of the cabinet to surge protector terminals 1 and 2 on the network side. **Do not** disconnect phone line from transmitter for the following verifications.

4. Make sure the broadcasting transmitter is connected to the Dummy Load and NOT to the Antenna. Turn ON the NWR1000 or NWR300 power switches for transmitters TXA and TXB.

5. Request the WFO transmit the alert tone from BMH for the purpose of verifying proper signal level on the audio program line and resulting modulation deviation.

6. At the audio input to the transmitter, measure the 1050 Hz warning alert tone level using an RMS AC voltmeter or AM-48. If it is not within the specified range of -9 to -12 dBm (0.275 to 0.195 Vrms), request a technician at the WFO check for 0 dB at the WFO DEMARC output (refer to Modification Note 80 Section 6.7) If the level at the WFO DEMARC is correct and the level of -9 to -12 dBm still cannot be obtained, check the phone line or perform transmitter troubleshooting procedures. Notify the appropriate NWS personnel.

7. Using a communications analyzer with an antenna on the input, measure the modulation on the digital display (4.5 kHz ± 0.5) for the transmitter connected to the external dummy load (TXA or TXB). Verify the sine wave on the CRT is symmetrical and is not distorted. If the deviation is not correct or the waveform displays distortion, perform transmitter troubleshooting.

NWRSAME Level Verification (Transmitter Site)

1. Request the WFO generate the NWRSAME signal from BMH.

2. At the audio input to the transmitter, measure the SAME Preamble level using an RMS AC voltmeter or AM-48. If it is not within the specified range of -14.5 dBm ± 0.5 dBm (0.146 ± 0.008), the BMH end-to-end alignment procedures must be performed. Notify the appropriate NWS personnel.

3.Using a communications analyzer with an antenna on the input, measure the modulation on the digital display (4.5 kHz ± 0.5) for the transmitter connected to the external dummy load (TXA or TXB) and record in Attachment E {8}. Verify the sine wave on the CRT is symmetrical and is not distorted. If the deviation is not correct or the waveform displays distortion, perform transmitter troubleshooting.

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Modulation Verification on Backup Transmitter (Transmitter Site)

1. Switch to the other transmitter (if applicable). The other transmitter should now be connected to the external dummy load (in replace of the antenna).

2. Request the WFO transmit the 1050 Hz alert tone from BMH for the purpose of verifying proper modulation deviation.

- Using a communications analyzer with an antenna on the input, measure the modulation on the digital display (4.5 kHz ± 0.5) for the transmitter connected to the external dummy load (TXA or TXB) and record in Attachment E {8}. Verify the sine wave on the CRT is symmetrical and is not distorted. If the deviation is not correct or the waveform displays distortion, perform transmitter troubleshooting.

3. Request the WFO to generate the NWRSAME signal from BMH for the purpose of verifying proper modulation deviation.

- Using a communications analyzer with an antenna on the input, measure the modulation on the digital display (4.5 kHz ± 0.5) for the transmitter connected to the external dummy load (TXA or TXB) and record in Attachment E {8}. Verify the sine wave on the CRT is symmetrical and is not distorted. If the deviation is not correct or the waveform displays distortion, perform transmitter troubleshooting.

Verify Proper Transmitter Transfer using BMH application; Final Transfer back to Appropriate Transmitter (Transmitter Site)

1. Verify transmitter is in normal mode of operation (Remote Control, Auto Switching Mode)

2. Call the WFO to coordinate the sending of transfer tones from the WFO to the transmitter site. Verify proper switching of the transmitters.

3. Have WFO send transfer tones to switch transmitter back to original online transmitter.

4. Once both transmitters operation has been verified to operate correctly, the transmitter can be returned to normal operation.

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Transmitter Audio Input Verification (WFO)

It is important that the output power levels from the BMH DAC ports be aligned accurately to provide proper stimulus to the NWR transmitters. Some DAC outputs will be directed to standard analog telephone lines, some to ultra-high frequency (UHF) links, and some to Voice over Internet Protocol (VOIP) channels or other transmission mechanisms, which will require differing output levels as defined in this section.

A separate alignment will be performed for each DAC Port used to drive NWR transmitters. DAC signal output levels will be adjusted separately for each of the following NWR signal types:

- Specific Area Message Encoding (SAME) Tones
- Alert Tone (1050 Hz)
- 1800 Hz Transmitter Switching Tone
- 2400 Hz Transmitter Switching Tone

DAC Outputs which are driving leased analog telephone lines shall be aligned to the following output signal levels (see Table A-2). Leased analog telephone lines have a 10 dB signal loss from the origin point, at the output of BMH, to the destination point, at the input to the NWR transmitter.

NWR Signal Type	Required BMH DAC Output Level
SAME Tones	-4.5 dBm ± 0.5 dB
Alert Tone (1050 Hz)	0 dBm ± 0.5 dB
1800 Hz Transmitter Switching Tone	0 dBm ± 0.5 dB
2400 Hz Transmitter Switching Tone	0 dBm ± 0.5 dB

Table A-2: DAC Output Levels for Analog Telephone Lines

DAC Outputs which are driving UHF links, Microwave links, or digital network-type connections to the transmitter site (e.g., T1, Digital Subscriber Line (DSL), 4G wireless) shall be aligned to the following output signal levels (see Table A-3). These levels are based on the assumption that the telecommunications circuits are lossless from origin to destination (0 dB signal loss). If there is a known signal loss for the specific link being used, the required DAC output levels will need to be adjusted accordingly to account for the loss in the link.

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NWR Signal Type	Required BMH DAC Output Level
SAME Tones	-14.5 dBm ± 0.5 dB
Alert Tone (1050 Hz)	-10 dBm ± 0.5 dB
1800 Hz Transmitter Switching Tone	-10 dBm ± 0.5 dB
2400 Hz Transmitter Switching Tone	-10 dBm ± 0.5 dB

Table A-3: DAC Output Levels for Lossless Telecommunication Links

1. From the BMH window, select **Transmitters** and click **Transmitter Alignment**. The *BMH Transmitter Group Alignment* window will appear as follows (see Figure A-16).

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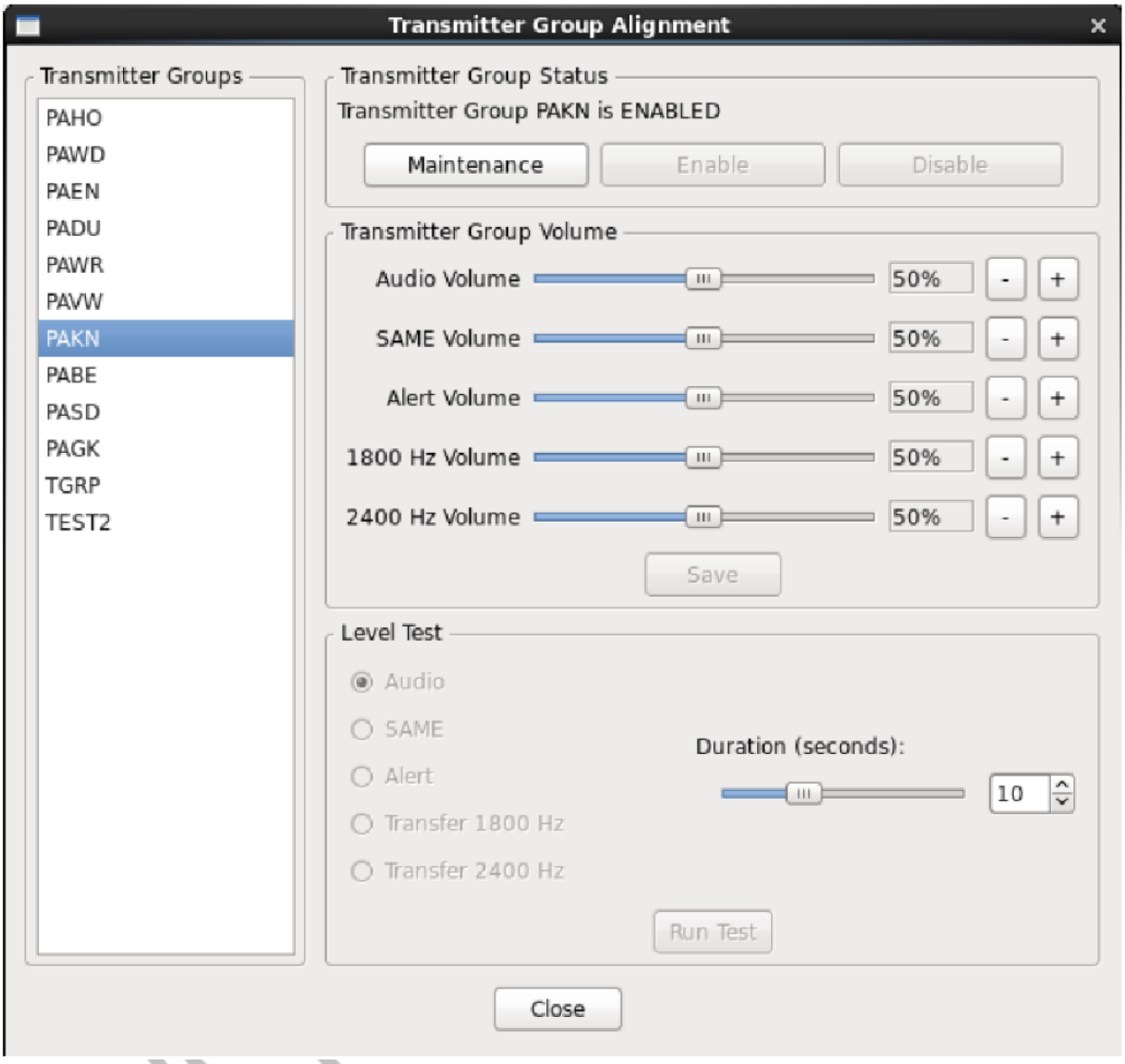


Figure A-16: Transmitter Group Alignment Window

2. Select the transmitter to be verified in the *Transmitter Groups* list in the *Transmitter Group Alignment* window.

NOTE: A mapping of the transmitter names to their associated DAC and Port numbers can be found on the *BMH Main Menu* screen.

3. Click the **Maintenance** button to take the selected transmitter offline in BMH and place the transmitter into Maintenance mode. The bottom part of the *Transmitter Alignment* window then activates (see Figure A-17).

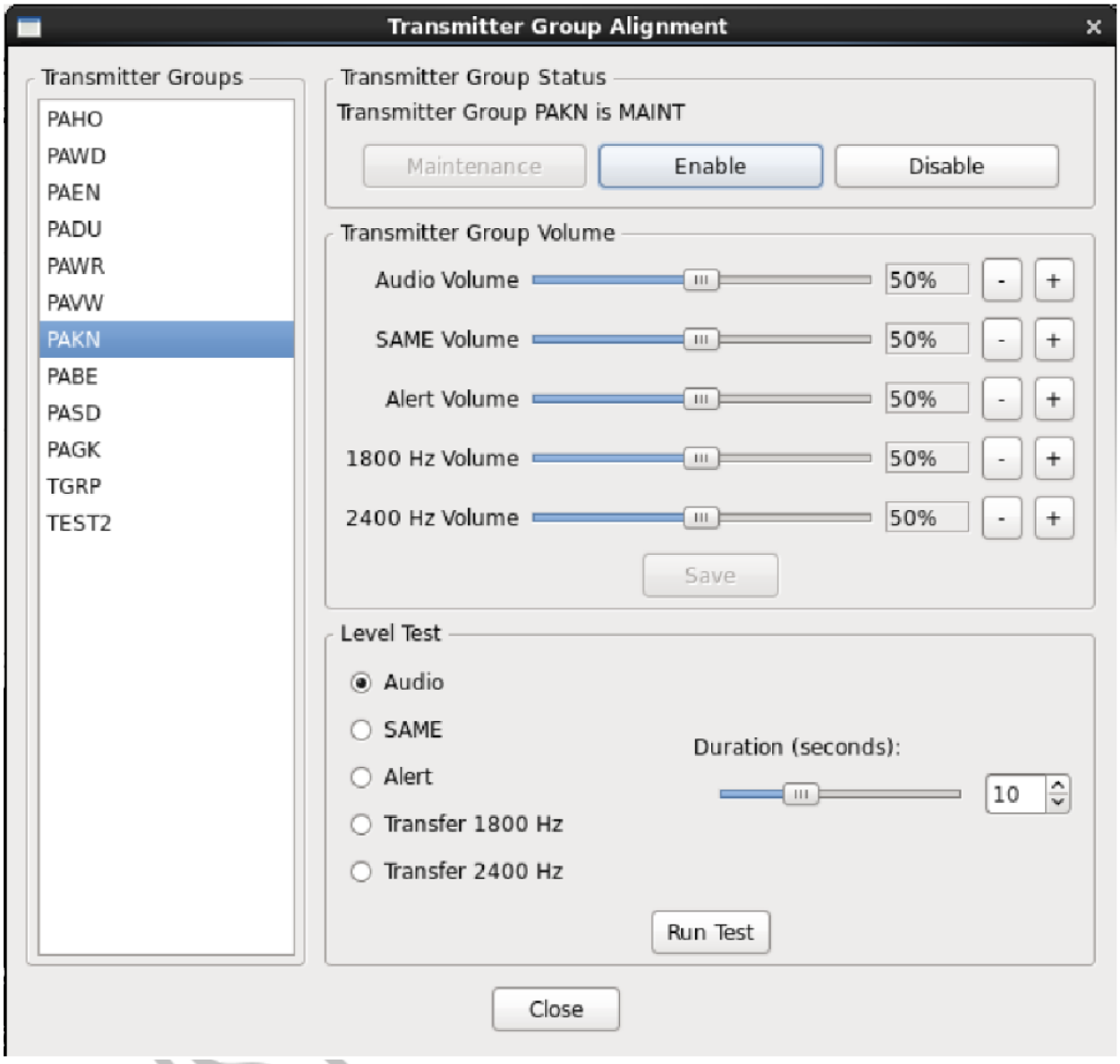


Figure A-17: Transmitter Group Alignment Window Active Status

NOTE: The Transmitter Group Volume slider bars should have already been aligned for the proper levels needed to feed the audio transmission lines from the WFO. If the levels received at the transmitter site are out of tolerance, verify the levels leaving the office. If the levels are good at the WFO, **DO NOT** compensate for an out of tolerance condition with the transmission line by adjusting the Transmitter Group Volume slider bars, fix the issue with the transmission line. If the levels are found to be out of tolerance at the output of the WFO, perform Transmitter Alignment procedure in Mod Note 80, section A.7.8

4. Select the **Alert** radio button in the Level Test portion of the window and click the **Run Test** button. To adjust the length of the tone emitted by the DAC, use the Duration adjustment in the window via the slider bar or the up or down arrows.

5. Select the **SAME** radio button in the Level Test portion of the window and click the **Run Test** button. To adjust the length of the tone emitted by the DAC, use the Duration adjustment in the window via the slider bar or the up or down arrows.

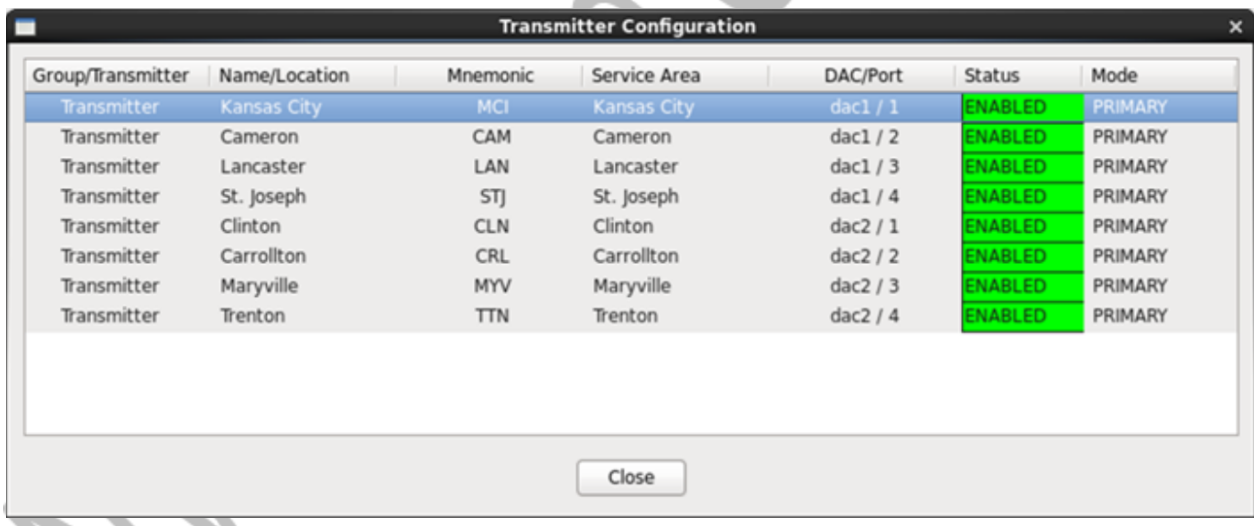
6. In the *Transmitter Group Status* portion at the top of the *Transmitter Groups Alignment* window Click the **Enable** button to take the selected transmitter out of Maintenance mode and resume normal transmitter broadcast feed to the transmitter.

7. Close the *Transmitter Groups Alignment* window by clicking the **Close** button located at the bottom of the *Transmitter Groups Alignment* window.

Verify Proper Transmitter Transfer using BMH application (WFO)

> **Note:** Verify with technician on site which transmitter is the online transmitter. It is possible the transmitter under test is not on the same transmitter (Primary or Secondary) as BMH indicates. If BMH has **Primary** selected and the transmitter under test is on transmitter #2 the transmitter under test will not switch the **Secondary** transmitter with Switching Tones. If BMH has **Secondary** selected and the transmitter under test is on transmitter #1 the transmitter under test will not switch the **Primary** transmitter with Switching Tones.

1. Open *Transmitter Configuration* window

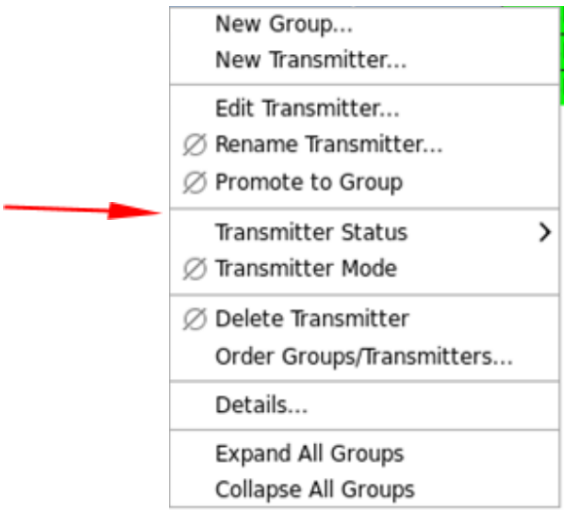


Group/Transmitter	Name/Location	Mnemonic	Service Area	DAC/Port	Status	Mode
Transmitter	Kansas City	MCI	Kansas City	dac1 / 1	ENABLED	PRIMARY
Transmitter	Cameron	CAM	Cameron	dac1 / 2	ENABLED	PRIMARY
Transmitter	Lancaster	LAN	Lancaster	dac1 / 3	ENABLED	PRIMARY
Transmitter	St. Joseph	STJ	St. Joseph	dac1 / 4	ENABLED	PRIMARY
Transmitter	Clinton	CLN	Clinton	dac2 / 1	ENABLED	PRIMARY
Transmitter	Carrollton	CRL	Carrollton	dac2 / 2	ENABLED	PRIMARY
Transmitter	Maryville	MYV	Maryville	dac2 / 3	ENABLED	PRIMARY
Transmitter	Trenton	TTN	Trenton	dac2 / 4	ENABLED	PRIMARY

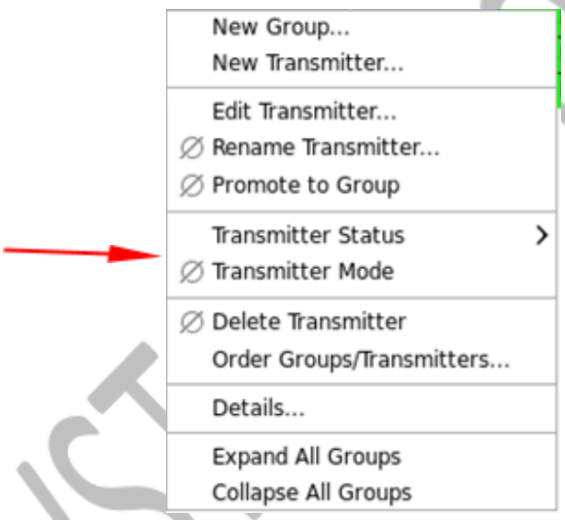
2. Right click on the transmitter under test.

> **Note:** The Transmitter Mode Option is disabled and will not be made available until the transmitter under test is disabled.

3. Click on *Transmitter Status* and **Disable** transmitter



4. Click on *Transmitter Mode* and select the radial button not selected, **Primary** or **Secondary**.



5. Verify with the technician on site, the transmitter switched from transmitter #1 to transmitter #2, if the **Secondary** radial button was selected. Verify with the technician on site, the transmitter switched from transmitter #2 to transmitter #1, if the **Primary** radial button was selected.

6. Switch the transmitter back to the desired transmitter to be online and verify with technician onsite, the correct transmitter is online and connected to the antenna.

7. Click on *Transmitter Status* and **Enable** transmitter.

8. Verify with Technician at the transmitter site no further switching or test tones are needed from the WFO.

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